

RESEARCH SKILLS

PNAP0164

WEEK 2 :

- FINDING RESEARCH PROJECTS
- DESIGN OF RESEARCH

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PART 1:

FINDING RESEARCH PROJECTS

“The seeds of great research lie in asking the right question.”
- Leedy & Ormrod (2021)

LEARNING OUTCOMES

By the end of this lesson, you will be able to:

1. **Identify** various types of research and their applications.
2. **Explain** the significance of research in academic and real-world contexts.
3. **Describe** the scientific method and its relevance to research.
4. **Outline** how research is conducted from start to finish.
5. **Illustrate** the research process using practical examples.
6. **Evaluate** criteria for what makes research “good” and credible.

INTRODUCTION

Every research journey starts with a problem or question worth exploring

Finding a research project means identifying a feasible, relevant, and impactful topic.

According to Kumar (2023), a good research idea originates from curiosity, existing knowledge gaps, and practical problems.

TYPES OF RESEARCH

By Purpose

- **Basic research** – expands knowledge without immediate application.
- **Applied research** – solves practical problems.

By Methodology

- **Quantitative** – numerical data, statistical analysis.
- **Qualitative** – explores meanings, experiences, narratives.
- **Mixed Methods** – combination of both.

By Timeframe

- **Cross-sectional** – one point in time.
- **Longitudinal** – over a period of time.

SIGNIFICANCE OF RESEARCH

Academic value :

Fills knowledge gaps,
builds theory.

Practical value :

Informs policies,
improves practices.

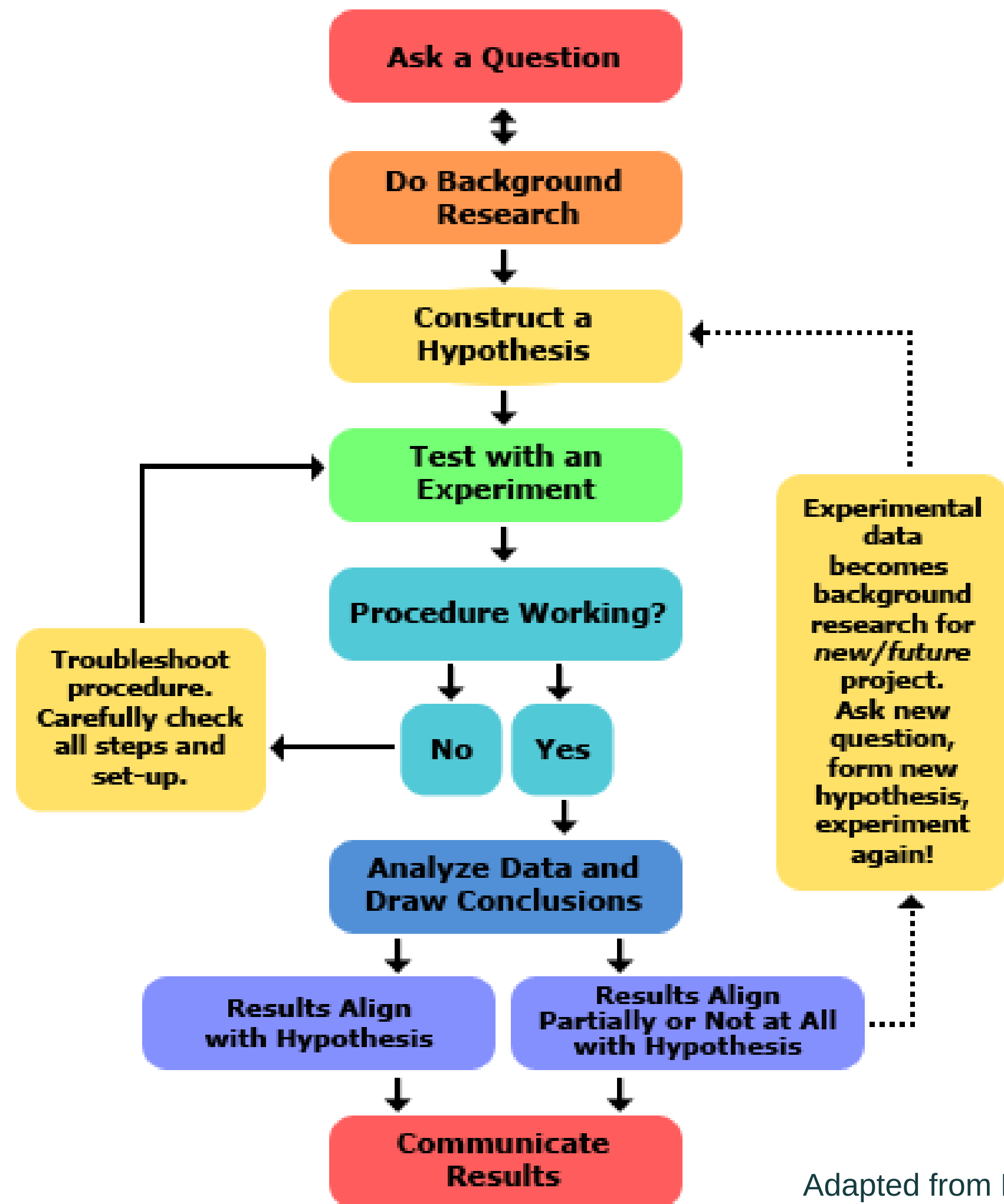
Personal value :

Develops critical thinking
& problem-solving skills.

Societal value :

Addresses pressing social,
economic, or
environmental issues.

SCIENTIFIC METHOD IN RESEARCH



Adapted from Kumar (2023).

STEPS OF THE SCIENTIFIC METHOD

1. Ask a Question

- The scientific method starts when you ask a question about something that you observe: How, What, When, Who, Which, Why, or Where?
- The question is something you can measure, preferably with a number.

2. Do Background Research

- Rather than starting from scratch in putting together a plan for answering your question, you can be a savvy researcher using library and Internet research to help you find the best way to do things and ensure that you don't repeat mistakes from the past.

3. Construct a Hypothesis

- A hypothesis is an educated guess about how things work. It is an attempt to answer your question with an explanation that can be tested.
- A good hypothesis allows you to then make a prediction: "If _____[I do this] _____, then _____[this]_____ will happen."
- State both your hypothesis and the resulting prediction you will be testing. Predictions must be easy to measure.

4. Test Your Hypothesis by Doing an Experiment

- Your experiment tests whether your prediction is accurate and thus your hypothesis is supported or not.
- Your experiments need to be a fair test. You conduct a fair test by making sure that you change only one factor at a time while keeping all other conditions the same.
- You should also repeat your experiments several times to make sure that the first results weren't just an accident.

5. Analyze Your Data and Draw a Conclusion

- Once your experiment is complete, you collect your measurements and analyze them to see if they support your hypothesis or not.
- Researchers often find that their predictions were not accurate and their hypothesis was not supported, and in such cases, they will communicate the results of their experiment and then go back and construct a new hypothesis and prediction based on the information they learned during their experiment.
- This starts much of the process of the research method over again. Even if they find that their hypothesis was supported, they may want to test it again in a new way.

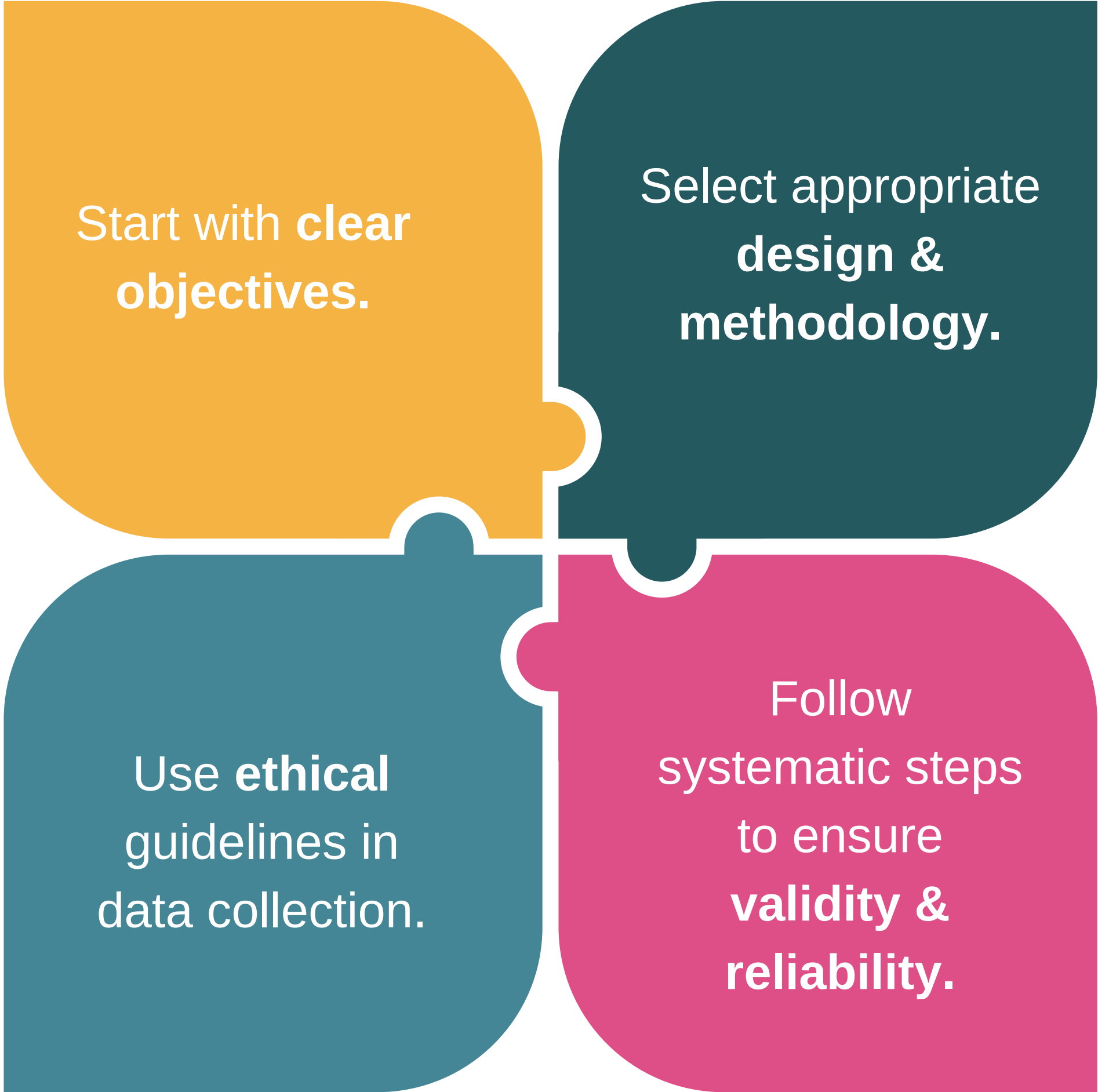
6. Communicate Your Results

- To complete your research, you will communicate your results to others in a final report or a display board.
- Professional researchers do almost the same thing by publishing their final report in a journal and/or by presenting their results during conference.



FUNDAMENTAL KNOWLEDGE OF RESEARCH

**Understanding the “what,” “why,” and “how” of
research before you actually do it.**



Start with clear
objectives.

Select appropriate
**design &
methodology.**

Use **ethical**
guidelines in
data collection.

Follow
systematic steps
to ensure
**validity &
reliability.**

HOW RESEARCH IS CONDUCTED

RESEARCH PROCESS

Conceptual phase

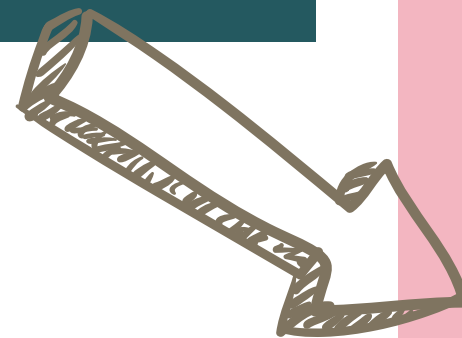
Problem
identification



Literature review



Hypothesis



Design phase

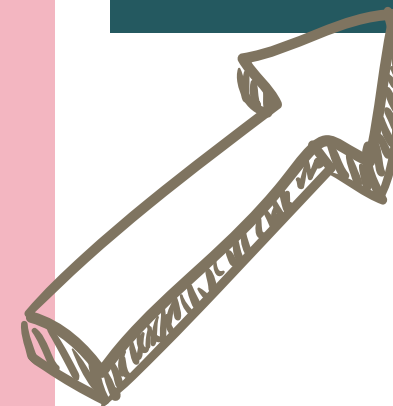
Choosing
research design



Sampling



Instruments

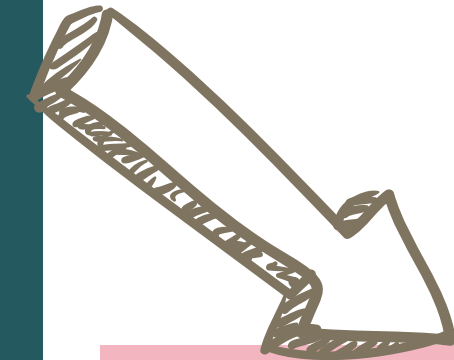


Empirical phase

Data collection



Data analysis



Dissemination phase

Reporting



Publishing



Applying findings

CRITERIA OF A GOOD RESEARCH

1. Clear purpose and focus
2. Well-grounded in theory/literature
3. Appropriate methodology
4. Ethical soundness
5. Reliability and validity
6. Original contribution

PART 2:

DESIGN OF RESEARCH

"A good research design provides the structure for answering research questions systematically." - O'LearyBell & Waters, 2024

LEARNING OUTCOMES

By the end of this lesson, you will be able to:

1. **Define** research design and its importance in the research process.
2. **Explain** key conceptual principles of research design.
3. **Apply** standard practice rules in designing a study.
4. **Outline** the main procedures for planning and executing a research design.

INTRODUCTION

Definition:

“A plan, structure, and strategy for investigating a research problem” (Leedy & Ormrod, 2021).

Ensures alignment between research problem, questions, methods, and analysis.

Acts as a blueprint for the study (Bordens & Abbott, 2022).

CONCEPTUAL PRINCIPLES

1. Clarity of purpose – research must have a clearly defined aim.
2. Coherence – all components of the study fit together logically.
3. Validity – design should enable accurate measurement of intended variables.
4. Reliability – results should be replicable.
5. Ethical integrity – protecting participants and respecting data privacy.

PRACTICE RULES

Match design to research question:

Exploratory, descriptive, experimental, etc.

Ensure feasibility:

Time, cost, and skill set.

Plan for bias control:

Randomisation, blinding, consistent procedures.

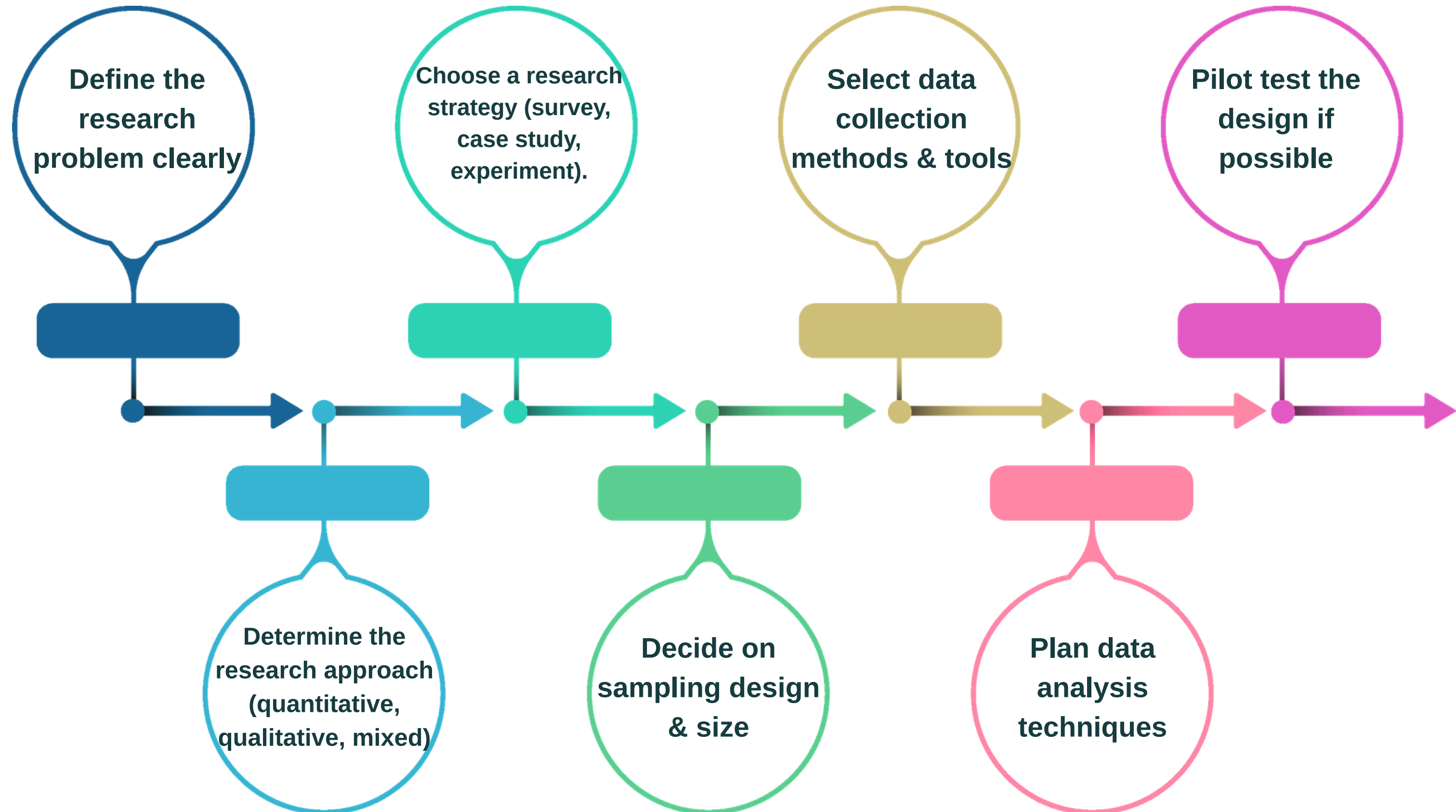
Integrate ethics early:

Informed consent, confidentiality.

Document decisions:

Maintain a design log.

PROCEDURES IN RESEARCH DESIGN



EXAMPLES

Research Problem:

Low engagement in online learning for gifted students

Design Type:

Mixed methods
(survey + focus groups)

Procedures:

1. Literature review
2. Questionnaire + interview design
3. Random sample selection
4. Pilot test
5. Data collection
6. Thematic & statistical analysis.

SUMMARY & DISCUSSION

SUMMARY :

A strong research design =
clear conceptual base +
practical feasibility +
systematic procedures.

DISCUSSION PROMPT :

Given a research topic, how would you ensure your design aligns with your research objectives?

CASE STUDY

TOPIC

A Technology-based Technopreneurship Education
Framework: Project-Based Learning Approach for
Gifted Students in the Sustainable Digital Economy

Background Problem

- Gifted secondary school students have high potential for innovation but lack structured, technology-driven entrepreneurship training aligned with sustainability goals.

Research Gap

- Existing studies on PBL, technopreneurship, and gifted education are often separate — rarely integrated into one framework tailored for the Malaysian gifted education context.

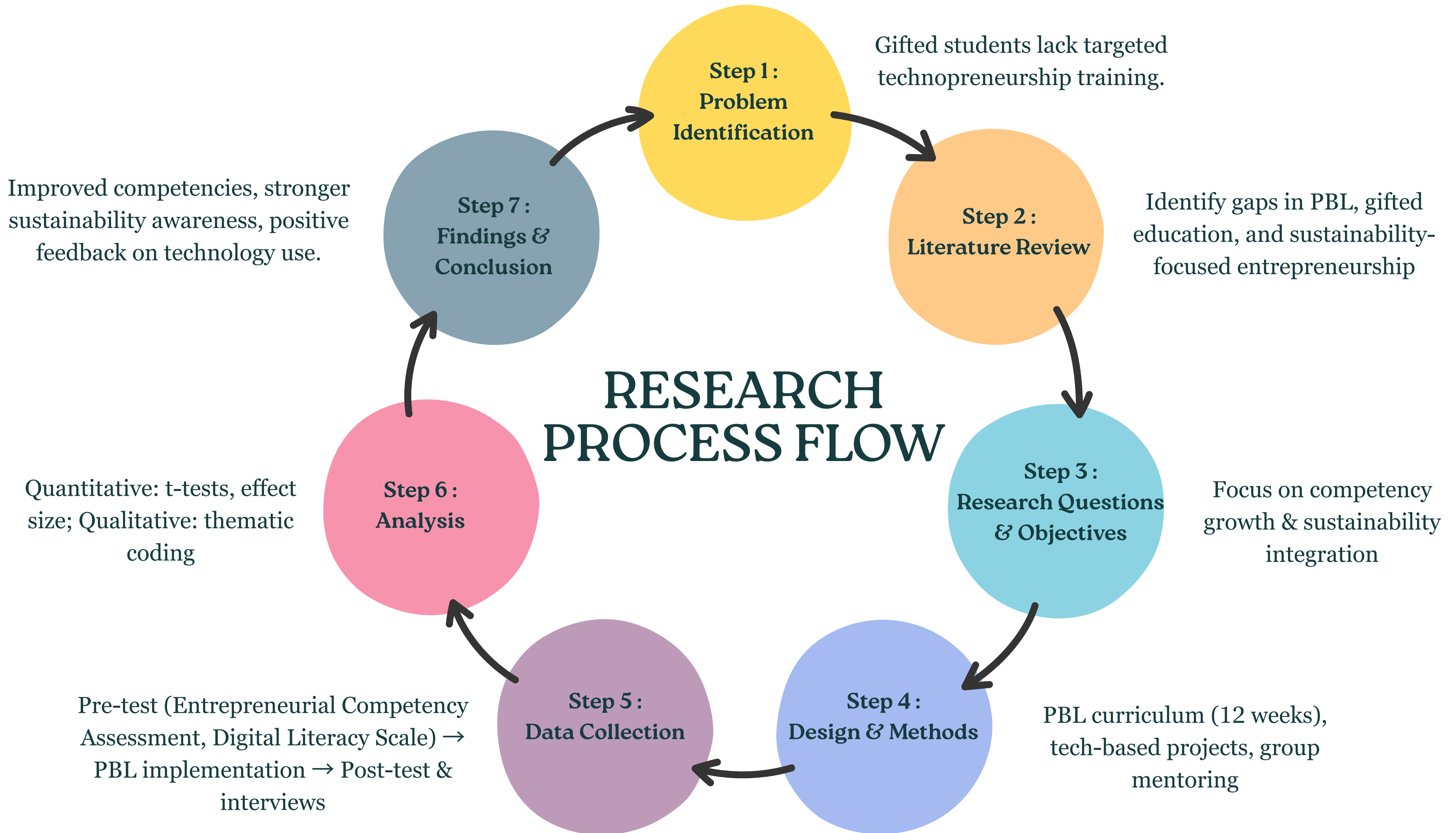
Objectives

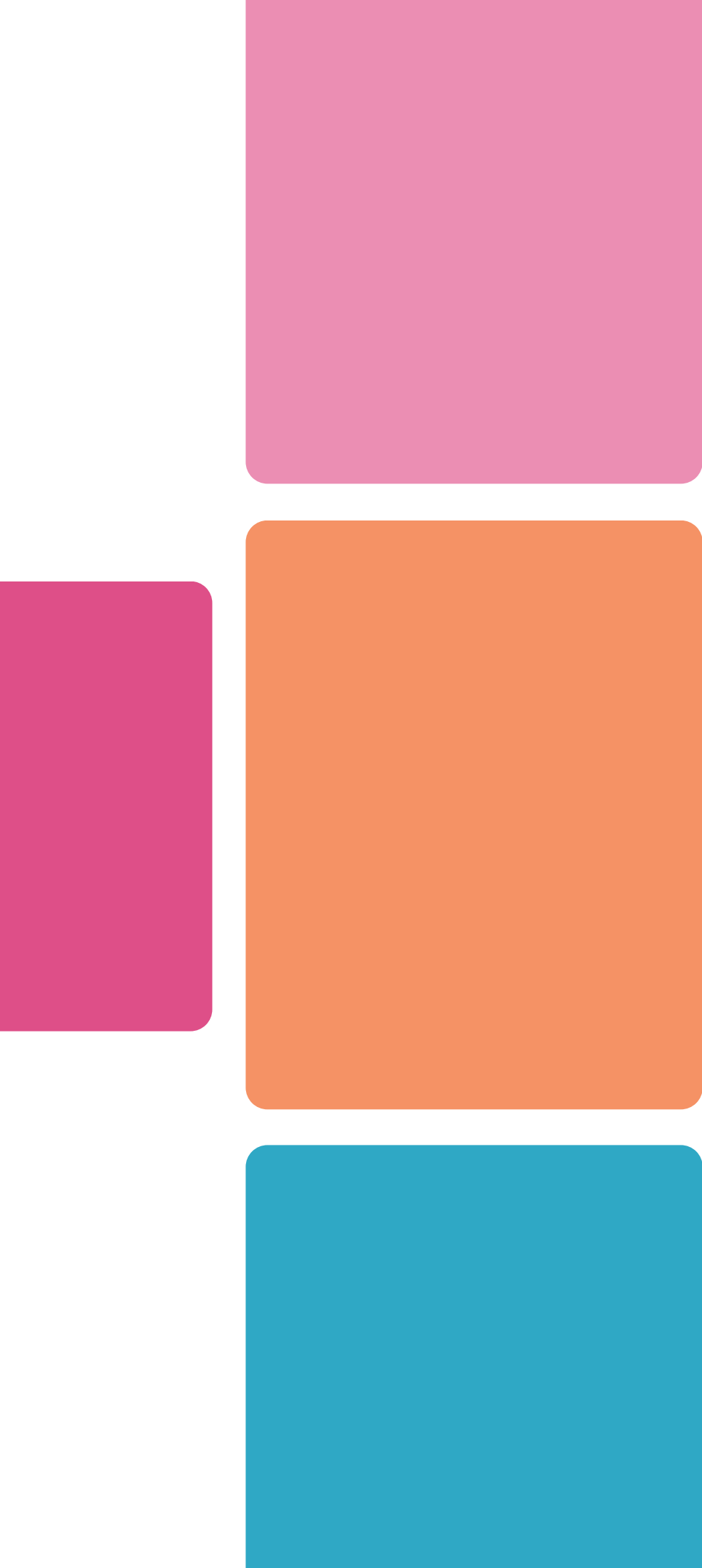
1. To design a technology-based PBL framework for gifted students.
2. To evaluate its impact on entrepreneurial competencies and sustainability awareness.
3. To explore students' experiences integrating technology into business projects addressing SDGs.

RESEARCH TYPE CLASSIFICATION

Aspect	Classification	Justification
Purpose	Applied Research	Focused on solving a real educational need.
Methodology	Mixed Methods	Combines quantitative skill assessments and qualitative reflections/interviews.
Timeframe	Longitudinal	Tracks development over two academic terms.
Design	Project-Based Learning Intervention	Students create tech-based business solutions for SDGs.

****Participants: 40 gifted students from Digital Innovator Programme.**



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THANK YOU